

REASONING SETTING (C, D, E)

● OpenThinker3-7B vs parent ◆ Qwen3-8B, thinking on vs off

CORPORA

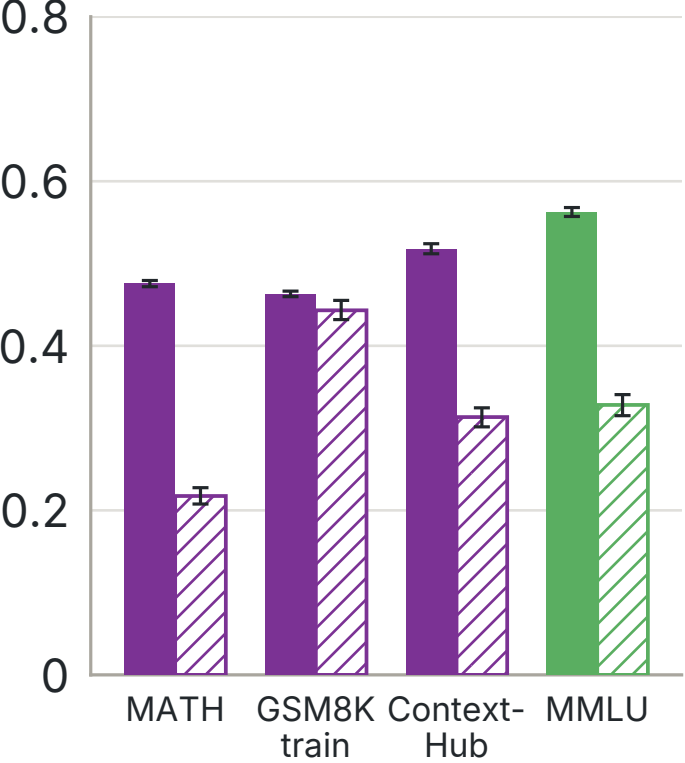
■ Needs-reasoning corpora ■ No-reasoning corpora

METRIC (A, B)

■ Held-out R^2 ▨ Answer retrieval (lift over chance)

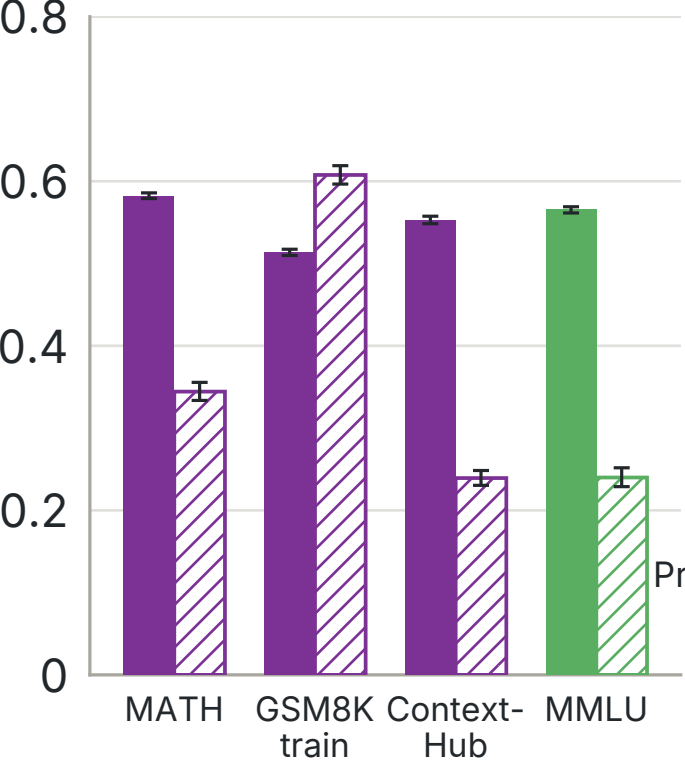
A · OPENTHINKER3-7B, CONTEXT → ANSWER

Needs-reasoning corpora are not less predictable than MMLU



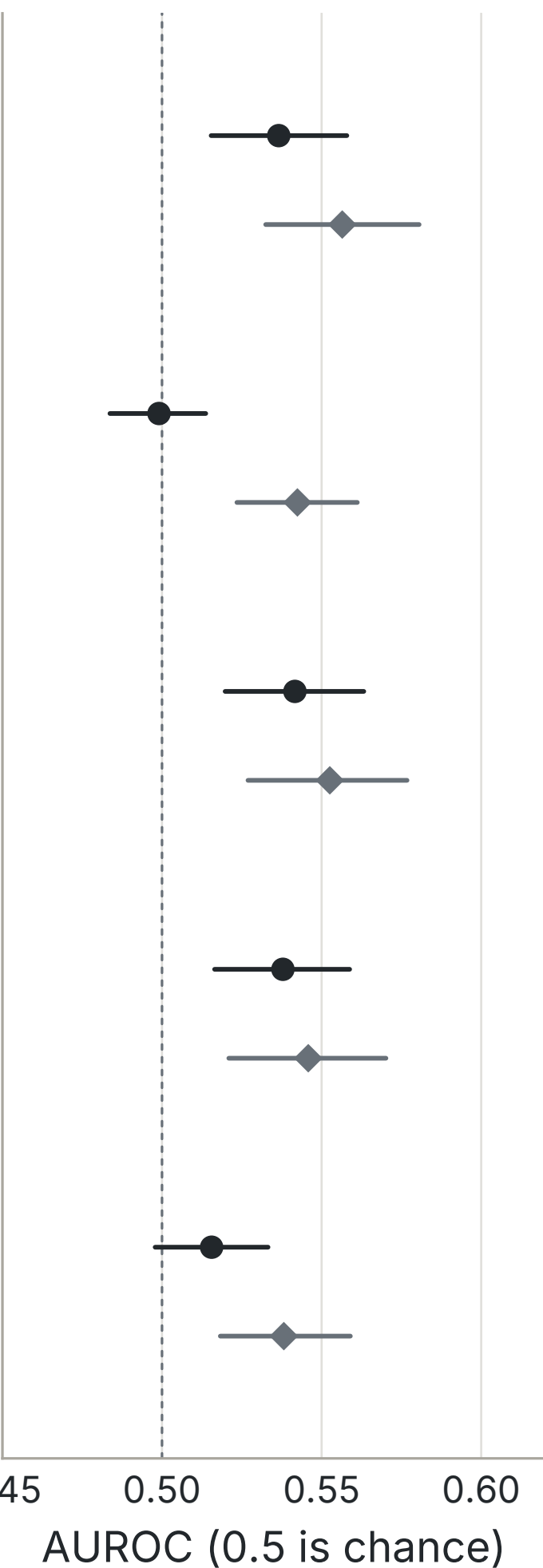
B · QWEN3-8B, CONTEXT → ANSWER

Same on Qwen3-8B



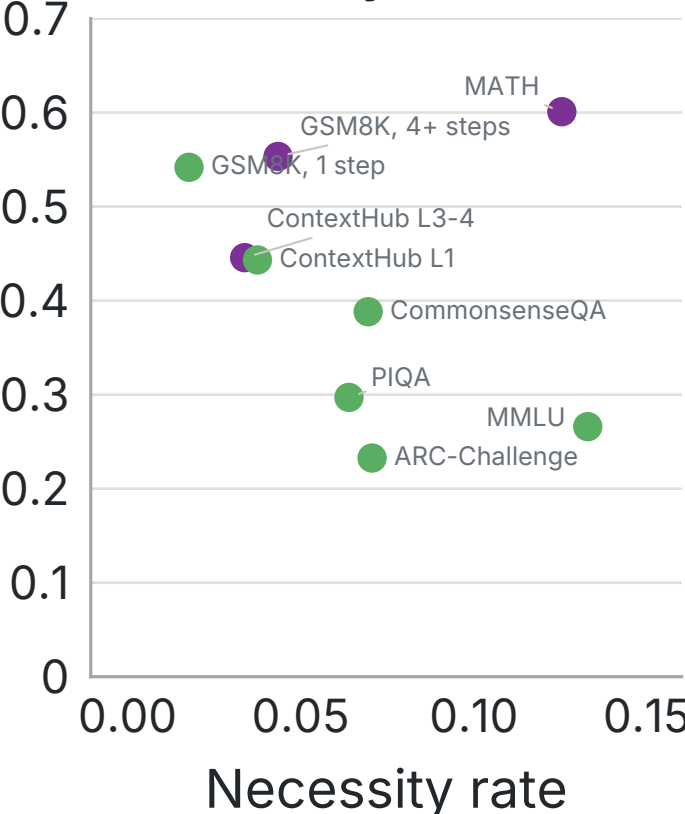
E · PER-QUESTION AUROC, 95% INTERVALS

No map score identifies which MATH questions need reasoning



C · OPENTHINKER3-7B VS PARENT

Gain follows corpus type, not necessity rate



D · QWEN3-8B, THINKING ON VS OFF

Same on the thinking toggle, at higher necessity rates

